

AYMANE AIT DADS

Developer Technology Intern, High-Performance Databases | GPU Optimization · Parallel Algorithms · LLM Fine-Tuning

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PROFESSIONAL SUMMARY

Engineered 5 sequential Blackwell GPU (sm_120, 95 GB VRAM) compatibility fixes enabling full 30B Mamba-Transformer MoE training on a novel architecture, directly demonstrating GPU architecture expertise and low-level CUDA debugging required by NVIDIA DevTech. Achieved top 10% public leaderboard (0.80/1.0) in the NVIDIA Nemotron Reasoning Challenge (\$106K+ prize pool) through 20+ single-variable ablation studies on LoRA adapter configurations across a 30B parameter model. Combines strong mathematical fundamentals with hands-on experience in parallel algorithm optimization, FP16/BF16 mixed precision, GPU profiling, Triton kernel debugging, and high-performance computing on latest-generation GPU architectures. Available Fall 2026; currently based in France and eligible to pursue US work authorization for internship placement.

CORE COMPETENCIES

GPU Architecture Optimization | Parallel Algorithms | High-Performance Computing | CUDA Debugging | Deep Learning Training | Ablation Studies | Performance Profiling | Mixed Precision Training

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Engineered an automated feature engineering and model evaluation pipeline on **100K+ fiber-optic network records**, reducing manual analysis time by approximately 60% for the network optimization team.
- Built a Random Forest classification model combined with K-Means clustering to map customer performance profiles across upload speed, latency, and jitter dimensions to **5 distinct commercial service tiers**.
- Developed a full clustering pipeline whose output was adopted directly by operations engineers, translating **thousands of network node classifications** into actionable maintenance recommendations.
- Delivered stakeholder presentations converting complex model outputs into maintenance strategy recommendations, communicating results to non-technical audiences across a cross-functional team using **Power BI dashboards**.

PROJECTS

NVIDIA Nemotron Reasoning Challenge - Kaggle Competition

Kaggle · In Progress June 2026 · Top 10% Public Leaderboard · \$106K+ Prize Pool

- Achieved **top 10% public leaderboard score of 0.80/1.0** (vs. 0.49 base, 0.86 best competitor) fine-tuning a 30B Mamba-Transformer MoE model (3.5B active parameters via MoE routing, 52 layers) with a LoRA adapter (r=32, ~888M trainable parameters) on 7,828 chain-of-thought reasoning examples using SFTTrainer with completion-only loss.
- Resolved **5 Blackwell GPU incompatibilities** (sm_120, 95 GB VRAM) in strict dependency order - including monkey-patching the CUDA caching allocator, stubbing mamba3/cutlass imports, disabling fast-path Mamba kernels, fixing Triton ptxas permissions, and replacing rmsnorm_fn with a pure PyTorch fallback - and conducted 20+ single-variable ablations confirming completion-only loss, packing strategy, and adapter size as critical performance factors.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton · Mamba

AI-Generated Image Detection - NTIRE 2026 @ CVPR

EURECOM ImSecu + NTIRE 2026 @ CVPR CodaBench · 2025 · Ranked 1st in EURECOM Class Leaderboard

- Ranked **1st on the EURECOM private class leaderboard (0.791 AUC)** and submitted independently to the NTIRE 2026 @ CVPR CodaBench international benchmark, fine-tuning CLIP ViT-L/14 (428M parameters) with a custom MLP head across 250K training images spanning 25+ generator types using dual learning rates and FP16 mixed precision.
- Identified that **1-epoch training (0.793 AUC) outperformed 4-epoch training (0.767 AUC)** through systematic ablation of unfreezing depth (4/6/8 transformer blocks), and applied a 3-model ensemble weighted by validation AUC with 10-view test-time augmentation to maximize out-of-distribution generalization.

PyTorch · OpenCLIP · ViT · AdamW · Kaggle T4 GPU

EDUCATION

Engineering Degree in Data Science (Master-Level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Cloud Computing, Statistics, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

LLM Fine-Tuning & GPU Engineering: LoRA / QLoRA | SFTTrainer | CUDA Debugging | Triton | FP16/BF16 Mixed Precision | Mamba-Transformer MoE | vLLM

ML / DL Frameworks & Computer Vision: PyTorch | Hugging Face Transformers | PEFT | TRL | Unsloth | OpenCLIP | Scikit-learn

Programming, Data & MLOps: Python (expert) | SQL | Bash | Pandas | NumPy | Git | Linux | Docker (basic)